



Qualification Pack



Refractory Brick Layer

QP Code: ISC/Q1201

Version: 3.0

NSQF Level: 3

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ISC/Q1201: Refractory Brick Layer

Brief Job Description

The individual in this role performs laying of right quality of refractory bricks with the right mortar in the designated areas in a steel plant, according to design specification.

Personal Attributes

The person should be patient, organised, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0009: Work effectively with others](#)
3. [ISC/N1201: Prepare for brick laying job](#)
4. [ISC/N1202: Lay refractory bricks according to design](#)
5. [ISC/N1211: Perform quality inspection of refractory materials](#)
6. [ISC/N1212: Carry out refractory debris segregation and disposal](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Refractory Maintenance
Country	India
NSQF Level	3
Credits	13



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7112.0300
Minimum Educational Qualification & Experience	10th Class with 6 Months of experience in relevant field OR 8th Class with 3 Years of experience in relevant field OR 5th Class with 7 Years of experience 7.5 years of experience in relevant field OR Certificate-NSQF (Level 2) with 3 Years of experience of experience in relevant field
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	3.0
Reference code on NQR	QG-03-IS-045362025-V2-ISSSC
NQR Version	1



Qualification Pack

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about following safety and adopting sustainable practices for optimising use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Emergencies, rescue and first aid procedures
- Health and hygiene
- Housekeeping and waste management
- Material and energy conservation

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** follow safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** use appropriate protective clothing/equipment for specific tasks and work
- PC4.** follow appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC5.** lift heavy objects safely using correct procedures
- PC6.** carry out routine check of the machine for identifying potential hazards
- PC7.** report any identified breaches in health, safety and security policies and procedures to the designated person

Emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC8.** use appropriate type of fire extinguisher
- PC9.** apply appropriate rescue techniques during fire hazard
- PC10.** provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.
- PC11.** follow emergency procedures such as raising alarm, safe evacuation etc.
- PC12.** attend safety training and fire drills to respond promptly during an emergency

Health and hygiene

To be competent, the user/individual on the job must be able to:

- PC13.** follow regular cleaning and disinfection practices at work place using appropriate techniques and materials
- PC14.** follow hand hygiene practices at work place using appropriate techniques and materials
- PC15.** report regarding the contagious illness of self or people in close contact
- PC16.** avoid contact with ill people and self-isolate in a similar situation



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Housekeeping and waste management

To be competent, the user/individual on the job must be able to:

- PC17.** follow the fundamentals of 5S for housekeeping
- PC18.** ensure good housekeeping in order to prevent hazards and accidents
- PC19.** store the material, tools and equipment in the correct location and in good condition
- PC20.** segregate waste into different categories
- PC21.** identify recyclable, non-recyclable and hazardous waste
- PC22.** dispose non-recyclable, recyclable and reusable waste appropriately at identified location

Material and energy conservation

To be competent, the user/individual on the job must be able to:

- PC23.** identify ways to optimize usage of material in various tasks/activities/processes
- PC24.** check for spills/leakages in various tasks/activities/processes
- PC25.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC26.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC27.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** preventative and remedial actions to be taken in case of exposure to toxic material
- KU7.** various types of fire extinguisher
- KU8.** various types of safety signs and their meaning
- KU9.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.
- KU10.** relevant standards, procedures and policies related to 5S followed in the company
- KU11.** the various materials used and their storage norms
- KU12.** efficient utilisation of material and water
- KU13.** basics of electricity and prevalent energy efficient devices
- KU14.** common practices of conserving electricity
- KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU16.** usage of different colors of dustbins



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KU17. waste management techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** ask for clarifications from superior about the job requirement
- GS4.** work with supervisors/team members to carry out work related tasks
- GS5.** complete tasks efficiently and accurately within stipulated time
- GS6.** inform/report to concerned person in case of any problem
- GS7.** make timely decisions for efficient utilization of resources
- GS8.** write reports such as accident report, in at least English/regional language
- GS9.** be punctual and utilize time efficiently



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	10	14	-	6
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	2	-	1
PC2. follow safe working practices while dealing with hazards to ensure safety of self and others	2	3	-	1
PC3. use appropriate protective clothing/equipment for specific tasks and work	1	2	-	1
PC4. follow appropriate safety practices while working in and around trenches, elevated places and confined areas	2	1	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	1
PC6. carry out routine check of the machine for identifying potential hazards	1	2	-	1
PC7. report any identified breaches in health, safety and security policies and procedures to the designated person	1	2	-	1
<i>Emergencies, rescue and first aid procedures</i>	6	9	-	5
PC8. use appropriate type of fire extinguisher	1	1	-	1
PC9. apply appropriate rescue techniques during fire hazard	1	2	-	1
PC10. provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.	2	2	-	1
PC11. follow emergency procedures such as raising alarm, safe evacuation etc.	1	2	-	1
PC12. attend safety training and fire drills to respond promptly during an emergency	1	2	-	1
<i>Health and hygiene</i>	2	6	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow regular cleaning and disinfection practices at work place using appropriate techniques and materials	1	2	-	1
PC14. follow hand hygiene practices at work place using appropriate techniques and materials	1	2	-	1
PC15. report regarding the contagious illness of self or people in close contact	-	1	-	-
PC16. avoid contact with ill people and self-isolate in a similar situation	-	1	-	-
<i>Housekeeping and waste management</i>	7	12	-	5
PC17. follow the fundamentals of 5S for housekeeping	2	3	-	2
PC18. ensure good housekeeping in order to prevent hazards and accidents	1	2	-	-
PC19. store the material, tools and equipment in the correct location and in good condition	1	2	-	-
PC20. segregate waste into different categories	1	2	-	1
PC21. identify recyclable, non-recyclable and hazardous waste	1	1	-	1
PC22. dispose non-recyclable, recyclable and reusable waste appropriately at identified location	1	2	-	1
<i>Material and energy conservation</i>	5	9	-	2
PC23. identify ways to optimize usage of material in various tasks/activities/processes	1	2	-	-
PC24. check for spills/leakages in various tasks/activities/processes	1	2	-	1
PC25. plug spills/leakages and escalate to appropriate authority if unable to rectify	1	2	-	1
PC26. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	1	-	-
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Cold Rolling Mill Operations, Electrical Maintenance, Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ISC/N0009: Work effectively with others

Description

This OS unit is about communicating with colleagues/superiors and others, either in own work group or in other work groups within organisation.

Scope

The scope covers the following :

- Communicate effectively with colleagues and others
- Interact with supervisor
- Follow appropriate behaviour at work place

Elements and Performance Criteria

Communicate effectively with colleagues and others

To be competent, the user/individual on the job must be able to:

- PC1.** coordinate with colleagues to share work, as per the workload in order to achieve team goals
- PC2.** maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written
- PC3.** adjust communication styles to reflect gender and persons with disability (PwD) sensitivity
- PC4.** respect all colleagues and co-workers
- PC5.** resolve conflicts by communicating with colleagues and other departments

Interact with supervisor

To be competent, the user/individual on the job must be able to:

- PC6.** identify work requirements by receiving instructions from reporting supervisor
- PC7.** escalate problems to supervisors that cannot be handled
- PC8.** report the completed work
- PC9.** interact with the reporting supervisor about any possible hazards and safety concerns

Follow appropriate behaviour at work place

To be competent, the user/individual on the job must be able to:

- PC10.** extend help to people with Disability (PwD) at workplace, if required
- PC11.** empathize with people with disability
- PC12.** adopt a gender neutral behavior
- PC13.** adopt responsible and disciplined behaviours at the workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor



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- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with colleagues and others</i>	13	20	-	9
PC1. coordinate with colleagues to share work, as per the workload in order to achieve team goals	3	5	-	2
PC2. maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written	5	7	-	3
PC3. adjust communication styles to reflect gender and persons with disability (PwD) sensitivity	3	4	-	2
PC4. respect all colleagues and co-workers	1	2	-	1
PC5. resolve conflicts by communicating with colleagues and other departments	1	2	-	1
<i>Interact with supervisor</i>	8	14	-	6
PC6. identify work requirements by receiving instructions from reporting supervisor	2	3	-	1
PC7. escalate problems to supervisors that cannot be handled	2	3	-	2
PC8. report the completed work	2	3	-	1
PC9. interact with the reporting supervisor about any possible hazards and safety concerns	2	5	-	2
<i>Follow appropriate behaviour at work place</i>	9	16	-	5
PC10. extend help to people with Disability (PwD) at workplace, if required	2	4	-	2
PC11. empathize with people with disability	2	4	-	1
PC12. adopt a gender neutral behavior	2	4	-	1
PC13. adopt responsible and disciplined behaviours at the workplace	3	4	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0009
NOS Name	Work effectively with others
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Operations, Coke Making, Mechanical Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1201: Prepare for brick laying job

Description

This OS unit is about preparing for the brick laying job as per the given work instructions and standards specified by the organisation.

Scope

The scope covers the following :

- Identify work requirements
- Prepare for brick laying job

Elements and Performance Criteria

Identify work requirements

To be competent, the user/individual on the job must be able to:

- PC1.** identify the work requirements by interpreting the drawing/sketch/specifications of the job and instructions received from supervisor
- PC2.** study the area where the bricking is to be performed
- PC3.** identify the tools, tackles & materials e.g. brick cutting machine, tape, power saw, hammer, spirit level, measuring tape, trowels, hand tools, jointing tools, wire-clamps, sledge hammer, chipping gun, chisel etc. required for the job
- PC4.** identify required quantity, sizes and numbers of the required raw materials e.g. different bricks, mortar powder, sand, clay etc.
- PC5.** select and arrange the right tools, tackles & materials as per the SOP and job requirements
- PC6.** report to stores / supervisor in case of non-availability of tools, tackles and material

Prepare for brick laying job

To be competent, the user/individual on the job must be able to:

- PC7.** use appropriate Personal Protective Equipment (PPE) for safe working at worksite
- PC8.** plan sequence of activities need to perform brick laying work
- PC9.** check the tools, tackles and material for any defects and that they are as per the required quality standards
- PC10.** prepare the mortar or/and the bonding material in the correct proportion by following the SOP
- PC11.** mix specified amounts of sand/ clay/ mortar powder with water in correct proportion to form refractory mixtures, or mortar for

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** basic terminologies of refractory bricking



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- KU3.** how to read and interpret basic drawing/specifications related to refractory bricking
- KU4.** Standard Operating Procedures (SOP) recommended by the manufacturer for using tools, and tackles required during work
- KU5.** characteristics of different types of bricks (composition, property & compatibility)
- KU6.** compatibility of the bricks/blocks with the setting material to be applied
- KU7.** methods to mix specified amount of sand/ clay/ mortar powder with water
- KU8.** safety requirements to be followed during the brick laying work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify work requirements</i>	17	23	-	11
PC1. identify the work requirements by interpreting the drawing/sketch/specifications of the job and instructions received from supervisor	2	3	-	1
PC2. study the area where the bricking is to be performed	2	3	-	2
PC3. identify the tools, tackles & materials e.g. brick cutting machine, tape, power saw, hammer, spirit level, measuring tape, trowels, hand tools, jointing tools, wire-clamps, sledge hammer, chipping gun, chisel etc. required for the job	5	4	-	3
PC4. identify required quantity, sizes and numbers of the required raw materials e.g. different bricks, mortar powder, sand, clay etc.	3	5	-	2
PC5. select and arrange the right tools, tackles & materials as per the SOP and job requirements	3	5	-	2
PC6. report to stores / supervisor in case of non-availability of tools, tackles and material	2	3	-	1
<i>Prepare for brick laying job</i>	13	27	-	9
PC7. use appropriate Personal Protective Equipment (PPE) for safe working at worksite	2	3	-	1
PC8. plan sequence of activities need to perform brick laying work	2	3	-	1
PC9. check the tools, tackles and material for any defects and that they are as per the required quality standards	3	7	-	3
PC10. prepare the mortar or/and the bonding material in the correct proportion by following the SOP	3	7	-	2
PC11. mix specified amounts of sand/ clay/ mortar powder with water in correct proportion to form refractory mixtures, or mortar for	3	7	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1201
NOS Name	Prepare for brick laying job
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Refractory Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1202: Lay refractory bricks according to design

Description

This OS unit is about laying the right quality of refractory bricks with the right mortar in the designated areas in a steel plant, according to design specifications.

Scope

The scope covers the following :

- Prepare surface for bricking
- Perform brick laying work

Elements and Performance Criteria

Prepare surface for bricking

To be competent, the user/individual on the job must be able to:

- PC1.** remove burned or damaged brick or mortar, using sledgehammer, crowbar, chipping gun, or chisel
- PC2.** clean the working surface and make it free from scale, dust, soot, or chips of brick and mortar, by using broom, wire brush, or scraper

Perform brick laying work

To be competent, the user/individual on the job must be able to:

- PC3.** identify the bricking requirements of a straight wall, corner wall, arch etc.
- PC4.** break or cut new bricks, tiles, or blocks into required size by using brick cutting machine, trowel edge, hammer, or power saw
- PC5.** follow safety precautions especially while handling heavy precast blocks/ slippery & smooth carbonaceous bricks
- PC6.** lay the bricks with the setting mixture on the designated areas requiring repair / replacement or new construction by following SOP
- PC7.** remove excess mortar by using trowels and hand tools
- PC8.** use jointing tools to give a sealed and uniform appearance finishing on mortar joints
- PC9.** ensure proper drying of the newly done surface
- PC10.** escalate the problems faced during brick laying activities to the supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** basic terminologies of refractory bricking
- KU3.** how to read and interpret basic drawing/specifications related to refractory bricking
- KU4.** Standard Operating Procedures (SOP) recommended by the manufacturer for using tools, and tackles required during work



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- KU5.** characteristics of different types of bricks (composition, property & compatibility)
- KU6.** compatibility of the bricks/blocks with the setting material to be applied
- KU7.** methods to break or cut new bricks, tiles, or blocks
- KU8.** how to build refractory arches where ever necessary
- KU9.** safety requirements to be followed during the brick laying work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read signals, work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organise work according to the work requirements
- GS8.** complete the assigned tasks with minimum supervisionhe type of concern
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare surface for bricking</i>	6	10	-	4
PC1. remove burned or damaged brick or mortar, using sledgehammer, crowbar, chipping gun, or chisel	3	5	-	2
PC2. clean the working surface and make it free from scale, dust, soot, or chips of brick and mortar, by using broom, wire brush, or scraper	3	5	-	2
<i>Perform brick laying work</i>	24	40	-	16
PC3. identify the bricking requirements of a straight wall, corner wall, arch etc.	2	3	-	2
PC4. break or cut new bricks, tiles, or blocks into required size by using brick cutting machine, trowel edge, hammer, or power saw	5	7	-	3
PC5. follow safety precautions especially while handling heavy precast blocks/ slippery & smooth carbonaceous bricks	2	3	-	1
PC6. lay the bricks with the setting mixture on the designated areas requiring repair / replacement or new construction by following SOP	5	11	-	4
PC7. remove excess mortar by using trowels and hand tools	3	5	-	2
PC8. use jointing tools to give a sealed and uniform appearance finishing on mortar joints	3	5	-	2
PC9. ensure proper drying of the newly done surface	2	3	-	1
PC10. escalate the problems faced during brick laying activities to the supervisor	2	3	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1202
NOS Name	Lay refractory bricks according to design
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Refractory Maintenance
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1211: Perform quality inspection of refractory materials

Description

This OS unit is about inspection of quality parameters (cracks, porosity, and shape deformities) of bricks before and after lying.

Scope

The scope covers the following :

- Inspect for cracks/warping
- Use basic tools for dimension check
- Document inspection results

Elements and Performance Criteria

Inspect for cracks/warping

To be competent, the user/individual on the job must be able to:

- PC1.** inspect refractory bricks and blocks visually for surface defects such as cracks, chipping, deformation, or discoloration before usage
- PC2.** check for proper labeling on incoming refractory materials to ensure correct usage
- PC3.** verify material storage conditions (dryness, stacking, temperature control) to ensure that materials have not degraded due to environmental exposure
- PC4.** conduct basic tap sound or manual knock tests to detect internal cracks or hollow sections, wherever applicable
- PC5.** identify mismatches in brick types (e.g., fireclay vs. high alumina) and ensure compatibility with job requirements and design drawings

Use basic tools for dimension check

To be competent, the user/individual on the job must be able to:

- PC6.** select appropriate measuring tools such as calipers, measuring tapes, or rulers based on the type and size of the refractory brick
- PC7.** check and ensure that the measuring tools are clean, calibrated, and in good working condition before use
- PC8.** measure the dimensions (length, width, thickness) of the refractory brick using calipers or measuring scale as per design specifications
- PC9.** compare the measured values against the required tolerances provided in the drawing or specification sheet
- PC10.** identify deviations or defects such as undersized, oversized, warped, or chipped bricks
- PC11.** mark the defective bricks and keep them aside

Document inspection results

To be competent, the user/individual on the job must be able to:

- PC12.** record the type and quantity of refractory materials inspected using standard log formats or checklists



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- PC13.** note observations such as visible cracks, spalling, warping, contamination, or dimensional discrepancies in refractory bricks
- PC14.** clearly mark defective or rejected materials and update the defect log accordingly
- PC15.** use basic tools (e.g., calipers, rulers) to verify and document physical measurements against specifications
- PC16.** submit the completed inspection documentation to the supervisor or quality control department as per SOP
- PC17.** maintain organized records of past inspection reports for future audits or reference
- PC18.** ensure that all documentation is completed accurately, legibly, and in compliance with organizational formats

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of refractory materials used in steel plants (e.g., fireclay, high alumina, silica, magnesia) and their respective applications
- KU2.** common physical defects in refractory materials such as cracks, chipping, spalling, warping, and discoloration, and their possible causes
- KU3.** importance of pre-use inspection of refractory materials to prevent installation failures or reduced service life
- KU4.** standard methods of visual inspection and basic manual testing (e.g., tap sound test) for internal defects
- KU5.** methods to identify and verify labeling information like batch number, material grade, and supplier name on incoming refractory stock
- KU6.** storage conditions required for different types of refractory materials including moisture control, stacking rules, and temperature requirements
- KU7.** basic measurement principles and how to use common measuring instruments like calipers, measuring tapes, and rulers
- KU8.** procedure to check the calibration and cleanliness of measuring instruments to ensure accurate readings
- KU9.** criteria for identifying non-conforming bricks based on dimensional deviation or physical damage
- KU10.** correct procedure for marking and segregating defective materials to prevent unintentional use
- KU11.** standard documentation formats (e.g., inspection checklist, defect log) used in quality control practices
- KU12.** protocols for submitting inspection data to supervisors or quality teams, including digital or paper-based systems
- KU13.** importance of maintaining accurate, clear, and up-to-date inspection records for traceability, quality audits, and compliance
- KU14.** relevant safety and handling procedures to follow while inspecting, measuring, and documenting refractory materials

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** ask for clarifications from superior about the job requirement
- GS4.** work with supervisors/team members to carry out work related tasks
- GS5.** complete tasks efficiently and accurately within stipulated time
- GS6.** inform/report to concerned person in case of any problem
- GS7.** make timely decisions for efficient utilization of resources
- GS8.** write reports such as accident report, in at least English/regional language
- GS9.** be punctual and utilize time efficiently



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect for cracks/warping</i>	10	20	-	6
PC1. inspect refractory bricks and blocks visually for surface defects such as cracks, chipping, deformation, or discoloration before usage	2	4	-	1
PC2. check for proper labeling on incoming refractory materials to ensure correct usage	2	4	-	1
PC3. verify material storage conditions (dryness, stacking, temperature control) to ensure that materials have not degraded due to environmental exposure	2	4	-	1
PC4. conduct basic tap sound or manual knock tests to detect internal cracks or hollow sections, wherever applicable	2	4	-	2
PC5. identify mismatches in brick types (e.g., fireclay vs. high alumina) and ensure compatibility with job requirements and design drawings	2	4	-	1
<i>Use basic tools for dimension check</i>	10	20	-	7
PC6. select appropriate measuring tools such as calipers, measuring tapes, or rulers based on the type and size of the refractory brick	2	4	-	1
PC7. check and ensure that the measuring tools are clean, calibrated, and in good working condition before use	1	3	-	1
PC8. measure the dimensions (length, width, thickness) of the refractory brick using calipers or measuring scale as per design specifications	2	3	-	2
PC9. compare the measured values against the required tolerances provided in the drawing or specification sheet	2	4	-	1
PC10. identify deviations or defects such as undersized, oversized, warped, or chipped bricks	2	3	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. mark the defective bricks and keep them aside	1	3	-	1
<i>Document inspection results</i>	10	10	-	7
PC12. record the type and quantity of refractory materials inspected using standard log formats or checklists	1	1	-	1
PC13. note observations such as visible cracks, spalling, warping, contamination, or dimensional discrepancies in refractory bricks	1	2	-	1
PC14. clearly mark defective or rejected materials and update the defect log accordingly	2	2	-	1
PC15. use basic tools (e.g., calipers, rulers) to verify and document physical measurements against specifications	2	1	-	1
PC16. submit the completed inspection documentation to the supervisor or quality control department as per SOP	1	2	-	1
PC17. maintain organized records of past inspection reports for future audits or reference	2	1	-	1
PC18. ensure that all documentation is completed accurately, legibly, and in compliance with organizational formats	1	1	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1211
NOS Name	Perform quality inspection of refractory materials
Sector	Iron and Steel
Sub-Sector	
Occupation	Refractory Maintenance
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1212: Carry out refractory debris segregation and disposal

Description

This OS unit is about segregation, labeling, and disposal of refractory debris in compliance with safety, environmental, and organizational waste management protocols.

Scope

The scope covers the following :

- Segregate recyclable and non-recyclable debris
- Label and document waste for compliance

Elements and Performance Criteria

Segregate recyclable and non-recyclable debris

To be competent, the user/individual on the job must be able to:

- PC1.** identify different types of refractory debris such as used firebricks, castable, insulating bricks, carbon blocks, and bonding material residues
- PC2.** distinguish between recyclable materials (e.g., unbroken firebricks, reusable precast blocks) and non-recyclable or hazardous waste (e.g., asbestos-contaminated material, carbon dust)
- PC3.** collect and segregate debris at the worksite using appropriate tools such as shovels, bins, and sorting trays, ensuring minimal dust generation
- PC4.** place recyclable and reusable debris in clearly labeled containers or designated collection areas as per organizational SOPs
- PC5.** ensure that non-recyclable or hazardous debris is stored in dedicated containers or bags compliant with workplace waste management protocols
- PC6.** wear appropriate PPE (gloves, respirator mask, safety shoes, etc.) while handling refractory debris to avoid exposure to fine particles or sharp edges
- PC7.** avoid mixing of different types of waste during collection and transportation within the site

Label and document waste for compliance

To be competent, the user/individual on the job must be able to:

- PC8.** label each category of waste using standardized tags or stickers, mentioning material type, date, and disposal classification (e.g., recyclable, hazardous)
- PC9.** document the quantity and type of waste generated in the prescribed waste tracking register or digital log
- PC10.** refer to material safety data sheets (MSDS) or organizational charts for proper classification of specific refractory debris
- PC11.** ensure that labels are legible, weather-resistant (if stored outdoors), and securely attached to waste containers or stockpiles
- PC12.** report unusual or excessive quantities of hazardous debris to the supervisor or environmental officer as per escalation procedures
- PC13.** follow company SOPs and legal/environmental norms (e.g., CPCB/SPCB guidelines) related to waste labeling and documentation



Qualification Pack

- PC14.** maintain organized records of waste generated and disposed of for future reference, audits, or inspections

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of refractory materials and debris such as firebricks, castables, insulating bricks, carbon blocks, and bonding material residues
- KU2.** distinction between recyclable/reusable materials (e.g., intact firebricks, clean precast shapes) and non-recyclable or hazardous waste (e.g., asbestos-laced insulation, carbon-rich debris)
- KU3.** environmental and health hazards associated with improper handling or mixing of refractory waste materials
- KU4.** purpose and methodology of safe collection and segregation of refractory debris using manual tools and equipment
- KU5.** use of appropriate Personal Protective Equipment (PPE) while handling, segregating, and transporting refractory waste, especially when dealing with dust, sharp edges, or hazardous materials
- KU6.** standard waste segregation practices followed at the workplace, including color-coded containers and dedicated bins for different types of waste
- KU7.** workplace protocols and SOPs for labeling, tagging, and tracking different types of refractory waste
- KU8.** importance of using weather-resistant, legible labels for containers, especially when stored outdoors or for prolonged periods
- KU9.** procedures for recording waste-related data in registers, forms, or digital platforms as per company protocol
- KU10.** escalation and reporting procedures in case of abnormal or excessive hazardous waste generation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** ask for clarifications from superior about the job requirement
- GS4.** work with supervisors/team members to carry out work related tasks
- GS5.** complete tasks efficiently and accurately within stipulated time
- GS6.** inform/report to concerned person in case of any problem
- GS7.** make timely decisions for efficient utilization of resources
- GS8.** write reports such as accident report, in at least English/regional language
- GS9.** be punctual and utilize time efficiently



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Segregate recyclable and non-recyclable debris</i>	15	25	-	10
PC1. identify different types of refractory debris such as used firebricks, castable, insulating bricks, carbon blocks, and bonding material residues	2	4	-	1
PC2. distinguish between recyclable materials (e.g., unbroken firebricks, reusable precast blocks) and non-recyclable or hazardous waste (e.g., asbestos-contaminated material, carbon dust)	2	3	-	2
PC3. collect and segregate debris at the worksite using appropriate tools such as shovels, bins, and sorting trays, ensuring minimal dust generation	2	4	-	2
PC4. place recyclable and reusable debris in clearly labeled containers or designated collection areas as per organizational SOPs	3	4	-	1
PC5. ensure that non-recyclable or hazardous debris is stored in dedicated containers or bags compliant with workplace waste management protocols	2	3	-	1
PC6. wear appropriate PPE (gloves, respirator mask, safety shoes, etc.) while handling refractory debris to avoid exposure to fine particles or sharp edges	2	4	-	2
PC7. avoid mixing of different types of waste during collection and transportation within the site	2	3	-	1
<i>Label and document waste for compliance</i>	15	25	-	10
PC8. label each category of waste using standardized tags or stickers, mentioning material type, date, and disposal classification (e.g., recyclable, hazardous)	2	4	-	1
PC9. document the quantity and type of waste generated in the prescribed waste tracking register or digital log	2	3	-	2
PC10. refer to material safety data sheets (MSDS) or organizational charts for proper classification of specific refractory debris	2	4	-	2



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure that labels are legible, weather-resistant (if stored outdoors), and securely attached to waste containers or stockpiles	3	4	-	1
PC12. report unusual or excessive quantities of hazardous debris to the supervisor or environmental officer as per escalation procedures	2	3	-	1
PC13. follow company SOPs and legal/environmental norms (e.g., CPCB/SPCB guidelines) related to waste labeling and documentation	2	4	-	2
PC14. maintain organized records of waste generated and disposed of for future reference, audits, or inspections	2	3	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1212
NOS Name	Carry out refractory debris segregation and disposal
Sector	Iron and Steel
Sub-Sector	
Occupation	Refractory Maintenance
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQF Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	30	50	0	20	100	10
ISC/N0009.Work effectively with others	30	50	0	20	100	10
ISC/N1201.Prepare for brick laying job	30	50	-	20	100	15
ISC/N1202.Lay refractory bricks according to design	30	50	-	20	100	15
ISC/N1211.Perform quality inspection of refractory materials	30	50	0	20	100	20
ISC/N1212.Carry out refractory debris segregation and disposal	30	50	0	20	100	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	200	330	-	120	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
5S	Technique of maintaining orderliness “Japanese terminology.
CP	Control Plan.
WI	Work Instructions.



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.